Report on "YGAME-D-25-00181"

Title: Preferences for Warning Signal Quality: Experimental Evidence

Summary

The revised manuscript is substantially improved. The authors collected an additional wave of data, expanded the treatment support, addressed the earlier boundary-value concern, and included new cross-task and mechanism analyses. These changes make the paper more credible than the original submission. However, several central issues remain unresolved. Most importantly, the definitions of false-positive and false-negative costs are still unclear and, in some places, incorrect. This affects the interpretation of Result 2 and Table 5, which are central to the paper's contribution. In addition, the theoretical appendix is not consistent with the revised main text. The main behavioral interpretation is interesting but should also be stated more cautiously.

Major Comments

Incorrect and Inconsistent Definitions of FP and FN Costs

The paper now emphasizes the distinction between false-positive/false-negative rates and false-positive/false-negative costs. This is a useful clarification. However, the definitions given in the model and in Section 5.2 are not correct. The paper defines FP and FN costs on page 6 as $\pi(1 - P_{01})c$ and $\pi P_{01}L$, respectively, and states that WTP is equally affected by both. There are two problems.

First, the FP cost definition is incorrect. The expression $\pi(1 - P_{01})c$ is the expected cost of true positive protection (correctly protecting in the adverse state), not the cost of false positives. The FP cost should be the expected cost of unnecessary protection when there is no adverse event: $(1 - \pi)P_{10}c$. This is consistent with equation (4), where the derivative of risk-neutral WTP with respect to the FP rate is $-(1 - \pi)c$.

Second, there is a subtlety in the FN cost definition. The paper defines it as $\pi P_{01}L$, which is the gross expected loss from missed detections. An alternative definition is $\pi P_{01}(L - c)$, which captures the incremental cost relative to the correct action: under a true positive signal the agent would have protected and paid c, so the net cost of missing the adverse state is $L - c$, not L. This alternative is more consistent with equation (5), where the derivative of WTP with respect to the FN rate is $-\pi(L - c)$.

This matters for Table 5 and the main regression. The dependent variable is WTP minus the risk-neutral value, and the independent variables are labeled FP and FN costs. The paper should explicitly define the variables used in the regressions, show how they map into the theoretical WTP formula (equation 3), and verify that the definitions do not affect the reported results.

The interpretation of Table 5 and Figure 5 should be stated more precisely

I found that the interpretation and conclusion of the paper are not clear and somewhat misleading. For instance, the abstract states that subjects "disproportionately reduce their demand for signals with high false negative rates for rare events." This characterization is imprecise and risks confusion with what the data actually show in subsection 5.2 (conditional on the results are correct).

Under the null hypothesis that subjects are risk-neutral expected-utility maximizers, the benchmark WTP already incorporates FP and FN costs, so all coefficients in the discrepancy regression (Table 5) should be zero. Positive coefficients indicate deviations from the benchmark: as error costs rise, subjects' WTP does not fall enough. The actual finding of the regression table is that subjects selectively under-react to the cost type that the theory identifies as dominant: for low priors, the coefficient on FP costs is large and significant while the coefficient on FN costs is near zero; for high priors, the pattern reverses, with FN costs generating the significant positive coefficient. The theory (equations 4–5) predicts that FP costs should be the primary concern for rare events, since $(1 - \pi)$ is large and most positive signals will be false alarms, while FN costs should dominate for frequent events, since π is large and missed detections become costly. This implies that subjects under-react to precisely the cost type that theory says should matter most:

- Low priors $\Rightarrow$ FP costs should dominate $\Rightarrow$ subjects fail to adjust for FP costs
- High priors $\Rightarrow$ FN costs should dominate $\Rightarrow$ subjects fail to adjust for FN costs

Figure 5 corroborates this from a different angle: when raw WTP is regressed on FP and FN rates by prior, the empirical sensitivity coefficients are roughly equal and flat across priors, whereas theory predicts sharp divergence. The finding is not that subjects overpenalize FN rates for rare events in absolute terms, but rather that they fail to differentially weight FP and FN rates as the prior shifts, producing relative over-sensitivity to FN (under-sensitivity to FP) for low priors and the reverse for high priors. I think this is stronger than saying simply that subjects are "asymmetrically under-responsive by prior," but weaker than saying that subjects generally fail to distinguish FP and FN errors. The evidence shows that subjects do not process FP and FN rates in the benchmark Bayesian/EU way. It does not uniquely identify the cognitive mechanism.

The abstract and conclusion should be rewritten to state this more precisely. The paper should also be consistent about whether claims refer to error rates, expected costs, or deviations from the risk-neutral benchmark. The paper would also benefit from using "false alarm" and "missed detection" alongside the FP/FN notation, especially when discussing costs. These terms make it immediately clear why low-prior environments make false alarms more important and high-prior environments make missed detections more important.

The Mechanism Claim Should Be Softened

The discussion in Section 5.3 is much improved. The authors now consider risk preferences, probability weighting, anchoring, non-instrumental information, and anticipation of biased protection decisions. The additional BE and IP analyses are also useful. However, I remain unconvinced that the data uniquely identify a failure to distinguish false-positive and false-negative errors. The evidence is interesting: some subjects report using the proportion of dishonest gremlins as a decision rule, and the BE/IP analyses show anomalous responses to FP and FN rates even when those rates should not affect the posterior given the observed signal (Tables 6–7).

However, the mechanism is not cleanly identified. The same pattern could reflect a broader total-error-rate heuristic, a "dishonest gremlin" heuristic induced by the experimental framing, probability weighting, noisy Bayesian reasoning, task complexity, or some combination. The authors themselves acknowledge that probability weighting may play a role for low priors. More importantly, the leap from "subjects react to irrelevant error rates in BE and IP tasks" to "subjects conflate FP and FN when valuing signals" requires more thorough discussion and the implicit assumption that the same cognitive failure operates across tasks.

A more defensible conclusion would be: under this experimental representation, many subjects appear to use a simplified total-error or total-dishonesty heuristic rather than separately processing how FP and FN rates interact with the prior probability. This interpretation fits the subject explanations and the BE/IP evidence better than the stronger claim that subjects generally cannot distinguish FP and FN errors. It is still an interesting finding.

Relatedly, Section 4 should not be framed only as reassurance that subjects understood the task. The BP, IP, and BE patterns are part of the behavioral phenomenon. The data suggest that subjects respond directionally to incentives and information, but also display substantial heterogeneity, mistakes, and simplified reasoning. That framing would make the transition to the WTP results more natural.

Minor Comments

Proposition 2. Ensure the wording explicitly states that the proposition concerns the *ratio* of sensitivities $(\partial b/\partial P_{01})/(\partial b/\partial P_{10})$, not absolute levels. A brief economic interpretation would help: compared to risk-neutral agents, risk-averse agents are relatively more intolerant of missed detections (false negatives) but more forgiving of false alarms (false positives).

Appendix B: Proof Numbering. The main text contains two propositions, but the appendix has three proofs (B.1–B.3). The proof for Proposition 2 (ratio of sensitivities) appears in B.3. Section B.2 appears to prove a result for imprudent agents that no longer corresponds

to a proposition in the main text. B.2 should be dropped or the main text should be updated to match.

Appendix B.3: Algebra Error. In equations (13) and (14), the denominator of the last expression is $(1-\pi)L$. It should be $(1-\pi)c$. This can be verified by simplifying the middle expression:

$$\frac{\pi}{1-\pi} \cdot \frac{u'(\xi_1)(L-c)}{u'(\xi_2)\,c} \;=\; \frac{\pi(L-c)}{(1-\pi)\,c} \cdot \frac{u'(\xi_1)}{u'(\xi_2)}.$$

Cross-Reference Error. Page 26 references "Appendix Table A10," but the appendix table with matching content (WTP deviations with IP bias controls) is numbered A8. The table numbering appears to be off.

Typos

- Page 7: "Equations 6-s 7" → "Equations 6–7."
- Page 15, footnote 10: missing comma between the two parameter estimates ($\hat{\pi} = 0.43$ $\hat{\theta} = 0.25$).
- Page 18: "the value for for a prior of 0.2" — duplicated word.
- Page 27: "the observed sensitivity patterns emerges" → "emerge."
- Page 27: "characteristcs" → "characteristics."